

PH 604: Optical Properties of Solids

Course Code	PH 604
Course Name	Optical Properties of Solids
L-T-P-C	3-0-0-3
Prerequisites	Mathematical Physics (PH511) or Mathematics-2 (EP301), Quantum Mechanics (PH513) or Quantum Mechanics and Applications (PH301), Electromagnetic Theory (PH521) or Foundation of electrodynamics (IC221), Condensed Matter Physics (PH523) or Solid State Physics (PH501)
Distribution	Elective
Approval	recommended by the 36th BoA (Sep 2020)

Course Contents

- **Introduction**
- Optical processes, the complex refractive index and dielectric constant, quantum theory of radiative transition. [3 Lectures]
- **Propagation of light in solid**
- Phenomenological models-Drude and Lorentz models. quantum mechanical description, linear response functions and Kramers–Kronig relations, dispersion, birefringence, optical anisotropy. [4 Lectures]
- **Absorption of light**
- Interband transitions, transition rate, absorption in direct and indirect semiconductor, spin-orbit coupling, indirect gaps, Urbach tails, Landau levels, Franz-Keldysh effect, and absorption spectra. [3 Lectures]
- **Excitons**
- Frenkel vs. Wannier excitons, optical selection rules, effect of Coulomb interaction on interband absorption, Franck-Condon approximation, Huang-Rhys model, Wannier exciton – LO phonon bound states. [3 Lectures]
- **Luminescence**
- Emission from solids, Interband luminescence, photo and electro luminescence, photoluminescence spectroscopy. [3 Lectures]
- **Quantum structures**
- Low dimensional materials and their electronic structures, absorption of quantum well, quantum confined Stark-effect, photoluminescence, optical properties of quantum dots, recent advancement in confined optical materials like zero and two-dimensional materials. [6 Lectures]
- **Plasmonic systems**
- Metals, doped semiconductors, free carrier absorption and plasmons, surface and slab plasmons, plasmons in metallic particles, negative refraction. [3 Lectures]
- **Light-phonon interactions**
- Infrared and Raman active phonons, Phonons absorption and reflectivity, polaritons, polarons, inelastic light scattering (Raman and Brillouin scattering), Feynman diagrams for light scattering [4 Lectures]
- **Impurity centers in semiconductors**
- Electronic spectrum of shallow donors, multiple valleys, valley-orbit coupling and acceptors, pseudospin-orbit coupling, impurity bands and metal-insulator transition, localized vibrational modes, LO modes bound to neutral impurities, lattice dynamics of isoelectronic impurities and mixed crystals. [5 Lectures]
- **Nonlinear optics**
- Optical nonlinearities, second order nonlinearities, third-order nonlinearities, optical Kerr effect, stimulated Raman scattering, generation and detection of terahertz radiation and recent advancement in this field. [6 Lectures]

Text Books

1. Mark Fox, **Optical Properties of Solids**, Oxford University Press, 2010.

References

1. Yutaka Toyozawa, **Optical Processes of Solids**, Oxford University Press, 2010.
2. Frederick Wooten, **Optical Properties of Solids**, Academic Press, 2013.
3. M. S. Dresselhaus, **Solid State Physics Part II: Optical Properties of Solids**, 2001.